Introduction

2. Data Analytics

Data analytics in higher education provides unique opportunities to examine, understand, and model pedagogical processes. Consequently, the methodologies and processes underpinning data analytics in higher education have led to distinguishing, highly correlative terms such as Learning Analytics (LA), Academic Analytics (AA), and Educational Data Mining (EDM), where the outcome of one may become the input of another (Nguyen et al., 2020).

2.1 Types of Data

2.1.1 Nominal Data

Nominal data is a type of qualitative data which groups variables into categories. You can think of these categories as nouns or labels; they are purely descriptive, they don't have any quantitative or numeric value, and the various categories cannot be placed into any kind of meaningful order or hierarchy (Stevens, 2022).

Nominal Data			
Marital status	**Course**	**Nationality**	**Gender**
single	Communication Design	Portuguese	Male
married	Veterinary Nursing	Turkish	Female
widower	Informatics Engineering	German	Female

Figure 1: Nominal Data Example

2.1.2 Ordinal Data

Ordinal data classifies data while introducing an order, or ranking. For instance, measuring economic status using the hierarchy: 'wealthy', 'middle income' or 'poor.' However, there is no clearly defined interval between these categories (Hillier, 2023).

Ordinal Data		
Application Order	**Mother's qualification**	**Father's Qualification**
1st choice	Doctorate	Higher education - degree (1st cycle)
2nd choice	Master's	Bachelor's Degree

Figure 2: Ordinal Data Example

2.1.3 Interval/Ratio

Interval data are a level of measurement that involves data that's naturally quantitative (is usually measured in numbers). Specifically, interval data has an order (like ordinal data), plus the spaces between measurement points are equal (unlike ordinal data). While ratio data is similar to interval data but the zero value implies absence of it (Jansen, 2020).

Interval/Ratio		
Units enrolled	Unemployment Rate	Inflation Rate
15	30%	2%
17	25%	3%
18	20%	4%

Figure 3: Interval/Ratio Data Example

2.2 Types of Data Analytics

Data analytics is the practice of using analytics tools to derive insights from datasets to inform decisions. With data analytics, organizations can improve decision-making, streamline operations, and increase revenue (Chen, 2024).

2.2.1 Descriptive Analytics

Using historical data analysis, descriptive analytics aims to comprehend past events. It entails compiling data, identifying patterns and gaining insights into or learning from past events (Yasar, 2024).

2.2.2 Diagnostics Analytics

Diagnostic analytics delves deeper into data to determine the reasons behind specific events. It involves identifying root causes, determining correlations and uncovering relationships between different variables (Yasar, 2024).

2.2.3 Predictive Analytics

Predictive analytics forecasts future patterns or events using statistical algorithms and historical data. It attempts to forecast future events and evaluate the probability of various scenarios. Predictive analytics works with algorithms and methodologies such as linear regression or logistic regression models (Yasar, 2024).

2.24 Prescriptive Analytics

Prescriptive analytics goes beyond predicting future outcomes to recommend specific actions for decision-making. It uses optimization and simulation techniques to determine the best course of action for desired outcomes (Yasar, 2024).

3. Impact of Data Analytics in the Education Industry

This part aims to identify three business problems concerning student performance and its impact on different contexts through the application of data analytics. By using data-driven analytics, educational institutions can affirm these problems and make calculated, well-informed decisions for the sake of the students, the institution.

3.1 Problem 1: Enhance Student outcomes

3.2 Early identification of At Risk Students / Predicting Student Performance

At university level studies, the students usually abandon some subjects in which they were enrolled in for the academic year. Students would initiate this dynamic by skipping, or not attending, lectures, sometimes due to neglect or carelessness, while at other times as a result of their lack of understanding of the subject. Consequently, they would lose track of the class agenda followed by the professor. Yet, when they would attend after some time of absence, they would discover, upon their return, that they would get discouraged by their lack of understanding, which in turn would decide not to return to lectures, and study on their own. However, some of these students do fail to turn up for their final exams, while the fail rate of those who actually take the exams are high. The bigger problem occurs when the events described happen often to a student, a recurrence that could constitute the beginning of dropouts from college. This is a topic of concern in the universities, especially in the early grades. Dropouts by college students is a hot topic at present, there are many analysis carried out on the subject, trying to find the causes of it, whether it is social, family related, organizational, personal (M. Gascueña & Guadalupe García, 2012). Academic failure increased the likelihood of course attrition by 4.2 times. The students who failed and persisted attributed academic failure to a confluence of dispositional, situational, and institutional factors. There was a compounded effect of academic failure on already-vulnerable students resulting in strong negative emotions (Ajjawi et al., 2020).

In today's educational landscape, accurately predicting academic performance remains a critical challenge. Traditional methods often rely on limited data sources and fail to capture the complexity of factors influencing student success (Bagade, 2024).

Using Predictive and Prescriptive Analytics would be the most ideal approach to properly study these nominal, ordinal, interval, and ratio variables like ‘grades’, ‘ethnicity’, ‘study time’, etc.. These analysis would provide various information, from determining which students might need school help to get back on their feet, and what these students could do to improve their grades. Through building models for early prediction of study outcomes, then evaluating the models using cross-validation and misclassification errors to decide which model which model outperforms other models in terms of classification accuracy, we can present predictive results that can be easily understood by students, teachers, and academic staff (Kovačić & Green, 2010). It is crucial to identify at-risk students as early as possible during a course. With the shortcomings of current academic early warning systems it is important to develop prediction models with high accuracy for at-risk students (Marbouti et al., 2016).

3.3 Improving graduation rates

3.4 Improve Student Retention Strategies

Student retention is an important issue for higher education institutions as withdrawals from higher education prior to degree completion remain at about 30 % in the member countries of the Organisation for Economic Cooperation and Development (Mah, 2016). Early attrition, particularly within the first year of study, is attributed to a variety of reasons including students struggling with the transition to university, boredom, lack of interest, inadequate preparation, insufficient academic and social engagement, negotiating an appropriate balance between study-loads and personal commitments, and dissatisfaction with the course or university (Coates, 2014; Hobsons, 2014, as cited in Seidel & Kutieleh, 2017). This causes students to withdraw from their classes or studies.

Thus, higher education institutions have recently gained interest in collecting and mining these dynamic student data with learning analytics to gain insight into learners' experiences and to predict and optimize learning processes (Ferguson, 2012; Fiaidhi, 2014; Ifenthaler, 2015; Ifenthaler et al., 2014; Long and Siemens, 2011, as cited in Mah, 2016). Furthermore, digital badges are a relatively new technology in educational settings for representing learners' achievements, knowledge, skills, and competencies in formal and informal learning environments (EDUCAUSE, 2012; Gibson et al., 2013; Ifenthaler et al., 2016, as cited in Mah, 2016). Student retention focuses strongly on the student (understandable of course). Studies are interested in testing hypotheses by either exploring or explaining the perspective of the student. There are some examples (c.f. Thomas, 2002) of studies straying from this and instead looking at the perspective of the institution and how that affects students. In this regard, by u diagnostic, predictive and prescriptive analytics would be the most effective approach, identifying what is the probable cause, identifying student attrition, as well as prescribing ways to reduce student attrition. By analyzing student information such as student's background (socio-economic), individual attributes (age, employment history, family situation, etc.), pre-college schooling (where they attended), and even residence (on or off campus). Also, the level of commitment, academic and social integration shown by the student were other factors used to quantify student retention risk (McCarthy, 2019).

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